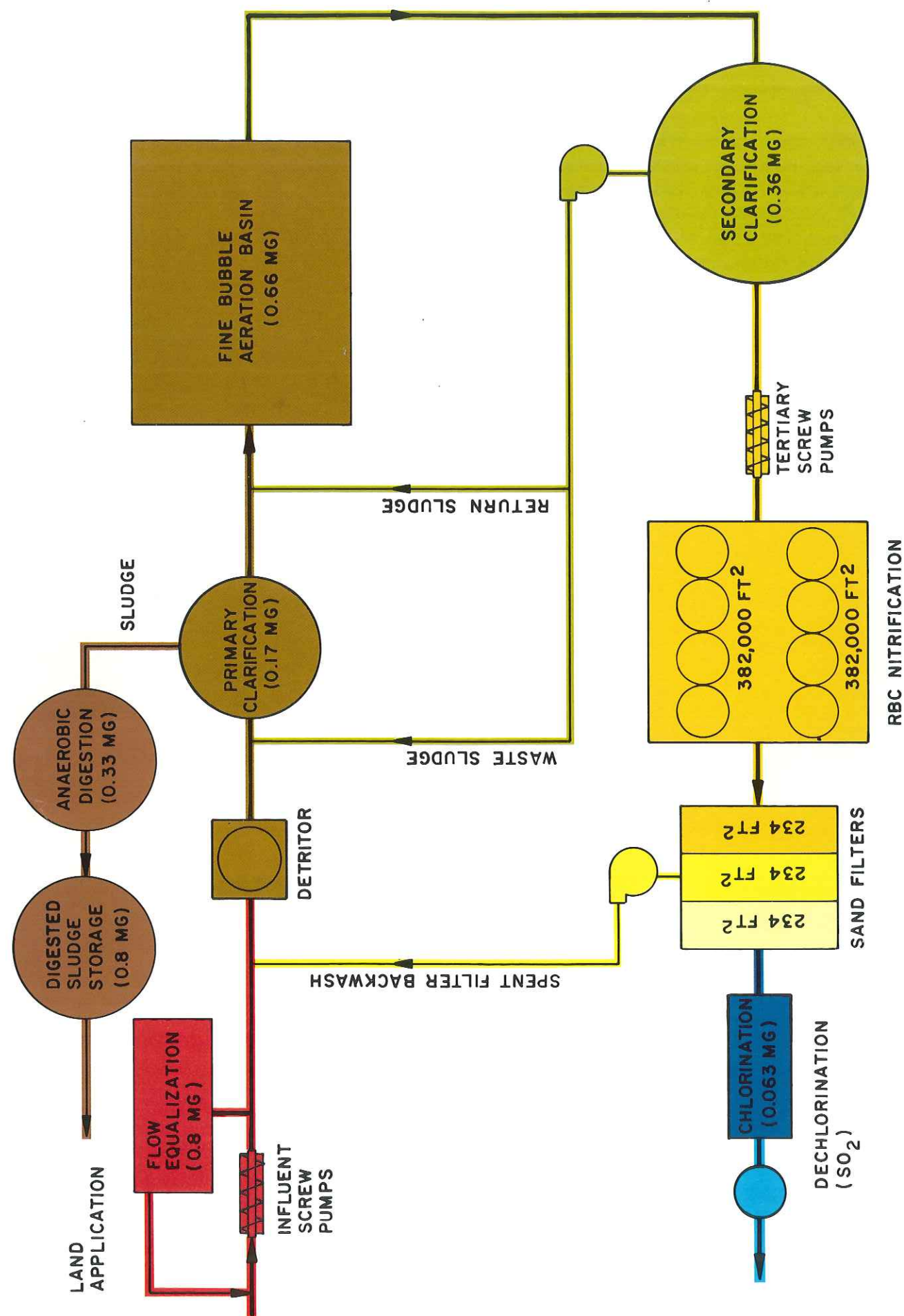




CADILLAC WASTEWATER TREATMENT





PROCESS FLOW DIAGRAM

The citizens, city officials, and employees of the City of Cadillac can be proud of their achievements in environmental protection and pollution control. The advanced wastewater treatment accomplished at the City's treatment facilities enhances the recreational use of the plant's receiving stream, the Clam River. The wastewater plant has proven itself more than capable of meeting or exceeding numerous stringent State and Federal treatment requirements with overall removal greater than 96 percent for biochemical oxygen demand, 96 percent for suspended solids, 93 percent for ammonia nitrogen and 90 percent for phosphorus. This plant is operated by the Wastewater Division of the Cadillac Utilities Department.

Cadillac's Progressive Wastewater Treatment Processes

Wastewater Treatment Processes:

- I. Original Plant: Constructed in 1908, consisted of a series of five filter beds that use slag for media.
- II. The next attempt to deal with sewage treatment began in 1923 and was completed in 1925. This facility utilized a septic tank with a capacity of 444,312 gallons; a 5,600 sq. ft. sludge bed and contact filter with 118,272 cu. ft. of slag filter media available. A 5-1/2 acre series of sand filters were constructed in 1939. Although adequate for its time, the old plant could not safely treat the increased flow by the 1960's.
- III. In March of 1962 construction began on a new "Activated Sludge" Treatment Plant. By May of 1963, sewage was flowing to the new plant. Several units, each with its separate waste removing processes, comprised the plant. These units can be grouped into three categories: primary or mechanical treatment with a grit removal system and two primary settling tanks; secondary or biological treatment with aeration tanks and two (secondary) settling tanks; and sludge digestion with two digesters and sludge drying beds.

Activated sludge is a mass of microscopic organisms which utilizes the sewage wastes as food, but must be fed constantly with air. These organisms will thrive and multiply, absorbing contaminants from the sewage. As the organisms grow they become heavy enough to be removed in the secondary settling tanks. Part of the sludge is returned to the sewage flow before the aeration process, to absorb more contaminants. The activated sludge is constantly growing and some must be wasted to the primary tanks; this and the other material settled in these tanks is pumped to the circular digestion tanks. Digestion is a different biological process. The sludge is kept warm and decomposes into methane gas and a black residue similar to the action in a septic tank. The gas produced is used to heat the sludge. After a period of time, the digested material is drained from the bottom of the digesters and either pumped to the sludge drying beds to dry and be used as a soil conditioner or injected into the soil as a fertilizer. The 1963 plant also included the addition of a biological laboratory with state-of-the-art equipment and procedures, many of which are still used today. Chlorination equipment which permits the killing of all harmful bacteria before the effluent is discharged into the river was also added at this time.

IV. 1974-75 Expansion - Phase I

The phosphorus removal facilities added to the existing primary and secondary treatment facilities consisted of chemical handling and feed equipment. Chemicals are added to the wastewater in the aeration tanks where they combine with phosphorus. The resultant precipitate is then settled along with sludge in the secondary clarifiers.

V. 1974-75 Expansion - Phase II

The installation of two tertiary screw pumps, tertiary building, rotating biological disc and rapid sand filters comprised the Phase II construction.

From the secondary clarifiers, wastewater flows to the tertiary screw pumps where it is pumped to the rotating biological discs for nitrification. The process uses a series of honeycomb discs mounted inside a horizontal tank. Wastewater flows through the tank so that only the lower portions of the discs are submerged.

Sand Filters

Effluent from the biological disc process is passed through a set of three sand filters. The filters are used to remove from the wastewater any suspended solids which may have passed through the final clarifiers and solids which have been produced from the biological growth on the rotating biological discs. Each filter provides 235 square feet of filter surface.

Chlorination and Dechlorination

As a final step, the treated wastewater is disinfected with chlorine in the chlorine contact tanks prior to discharge to the Clam River. A contact time of 50 minutes is provided in the tanks. Sulfur dioxide is used to remove unused chlorine from the final effluent to maintain chlorine residual below 0.036 ppm concentration.



Digesters and Sludge Holding Tanks

Sludge from primary and secondary treatment is pumped into digesters where anaerobic bacteria breakdown sludge into stable material. The digested sludge is stored in sludge storage tanks until it is transported to farmlands for recycling.

Laboratory

An Environmental Laboratory was built in conjunction with the plant expansion during 1989-90 project. This lab will analyze not only the wastewater through various process units, but also will analyze potable water for compliance with new drinking water act criteria and provide industrial waste analysis back-up for the plant lab.



As the discs rotate, sections of the discs are alternately exposed to wastewater and air and the discs become covered with a biological film. Microorganisms growing on the discs absorb oxygen from the air and use it to oxidize the ammonia in the wastewater to nitrates.

From the rotating biological discs, the wastewater is passed through a set of three rapid sand filters. The filters remove suspended solids which may have passed through the secondary clarification phase or which have been produced from the biological growths on the rotating discs. The effluent from the rapid sand filters is chlorinated to achieve disinfection prior to the discharge to the Clam River.

VI. 1989 - 1990 Expansion

Continued industrial, commercial, and residential growth began to approach the hydraulic capabilities of the still functioning 1963 designed plant. These near hydraulic overloads came in the form of peak flows during the day time hours. In order to compensate for these and anticipated future flows, City officials and engineering consultants considered the two options available to them. First, they could expand all phases of the treatment process, primary, secondary, and tertiary, to accommodate the diurnal flows, thus underutilizing the plant's capability a great deal of the time. The second option available was to equalize the flow into the plant. This has been accomplished with the addition of a 800,000 gal. equalization basin. All flow into the plant now passes through a diversion chamber connected to the equalization basin. If flows exceed a predetermined set point, the excess flows are diverted to the tank, which could be aerated to avoid septic conditions, then returned to the waste influent to be treated at low flow periods. When the equalization tank is empty it is cleaned daily by a series of four flush tanks mounted at the top of the structure. These flush tanks are unique as they require no mechanical equipment to work. As the tanks are filled with final effluent water the tanks become unstable just before overflowing thus emptying its contents. This rush of water produces such large mass impact that the sedimented debris lying on the tank floor is flushed back to the equalization tank drain to be treated in the wastewater flow.

The construction of a third influent screw pump and secondary clarifier was a welcome addition to plant personnel performing maintenance. In the past, any time one of these critical units was taken out of service for routine or emergency maintenance, the entire waste flow was being treated by one unit. The State also required the addition of dechlorination equipment to remove the toxic effect of chlorine on the receiving stream.

Odors are now practically eliminated with the addition of activated carbon odor control equipment being installed on the enclosed screw pumps and equalization basin.

Incorporated into the plant expansion was the construction and equipping of a new environmental laboratory facility adjacent to the existing Plant Control Laboratory. This facility will provide back-up of analytical procedures for industrial waste by the plant control lab and in addition, is equipped to provide monitoring of public water supplies for the new Safe Drinking Water Act regulations. This lab has the latest in analytical equipment for these purposes.

The City's unique advanced wastewater treatment facilities have been and continue to be widely acclaimed by the professionals. The 1974-75 Expansion won the Consulting Engineers Council of Michigan 1974 Eminent Conceptor Award for design excellence. The most recent 1990-91 construction earned the plant the 1990 Honors Award for engineering excellence presented by the Consulting Engineers Council of Michigan.

The plant is staffed by well trained and dedicated employees who take pride in operating and maintaining this up-to-date facility to its maximum potential.

THE PROCESS ILLUSTRATED

Screw Pumps And Equalization Basin

Three screw pumps, each with a capacity of 4.5 million gallons per day, are used to lift wastewater from the sewers. An equalization basin with a capacity of 800,000 gallons is used to store wastewater during the periods of high flows. The basin is drained during the lower flow and then automatically cleaned.



Detritor

The plant influent passes through the detritor where heavy inorganic solids such as sand and gravel are settled from the wastewater. Removal of inorganic solids early in the treatment process prevents excessive wear on subsequent treatment equipment. The detritor is 14 feet in diameter and has a detention time of about one minute.



Primary Tanks

Primary tanks are used to settle out suspended solids from the wastewater. This is accomplished by maintaining quiescent conditions in the tanks. Chemicals may be mixed with the wastewater prior to entering the primary tanks to aid in the removal of suspended solids and to remove phosphorus. Each tank is 40 feet in diameter with a 10.5 foot water depth. The detention time in the tanks is two hours.



Aeration Tanks

The settled primary tank effluent enters the aeration tanks where the wastewater is aerated until a large mass of solids form. The solids in the tanks are an aggregation of bacteria and other microorganisms that utilize constituents in the wastewater for food and in the process convert these constituents into settleable solids that may be removed in subsequent clarifier tanks. Chemicals are added to the wastewater in the aeration tanks where they combine with phosphorus and form a precipitate that also is settled out in the final clarifiers. Each aeration tank is 70 feet by 40 feet with a 15 foot water depth. The two tanks provide a detention time of 7.5 hours.

Secondary Clarifiers

Final clarifiers are used to settle out the suspended solids and phosphorus precipitate formed in the aeration tanks. Each tank is 45 feet in diameter with a nine foot water depth. The detention time in the clarifiers is 2.5 hours.

Screw Pumps

Clarified effluent from the final clarifiers is lifted by means of screw pumps to succeeding biological disc and sand filter units. Each pump is capable of pumping 4.5 million gallons per day.

Biological Discs

The biological disc units convert ammonia nitrogen to nitrate nitrogen. The process uses a series of honeycomb discs mounted inside tanks. A total of eight tanks are utilized. Wastewater flows through the tanks so that only the lower portion of the discs are submerged as the discs rotate. Sections of the discs are alternately exposed to wastewater and air, and the discs become covered with a biological film. Microorganisms growing on the discs absorb oxygen from the air and use it to oxidize the ammonia in the wastewater to nitrates. The nitrates can safely be released to the Clam River without depleting the oxygen supply in the river. A total of 764,000 square feet of disc surface area is provided.





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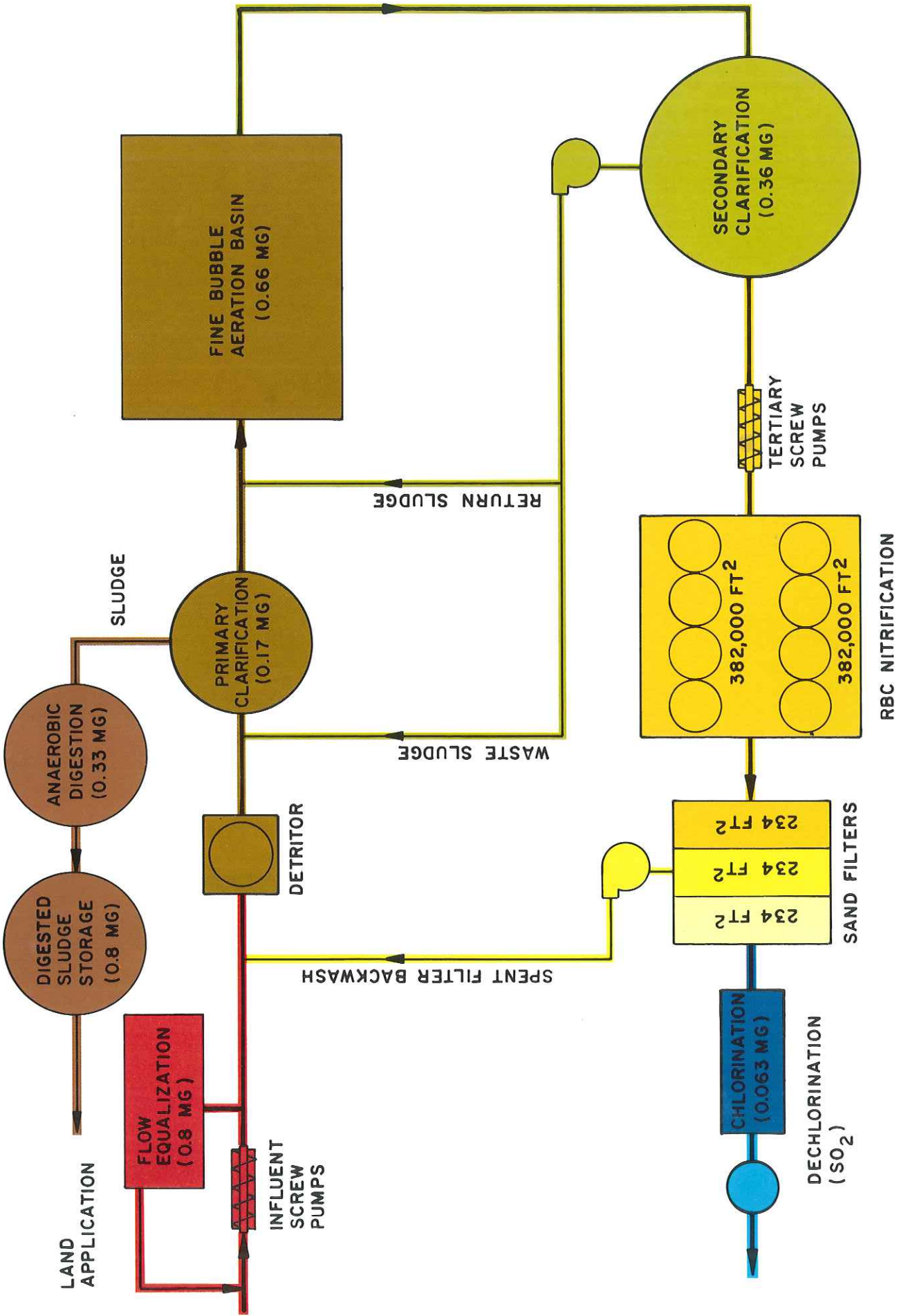
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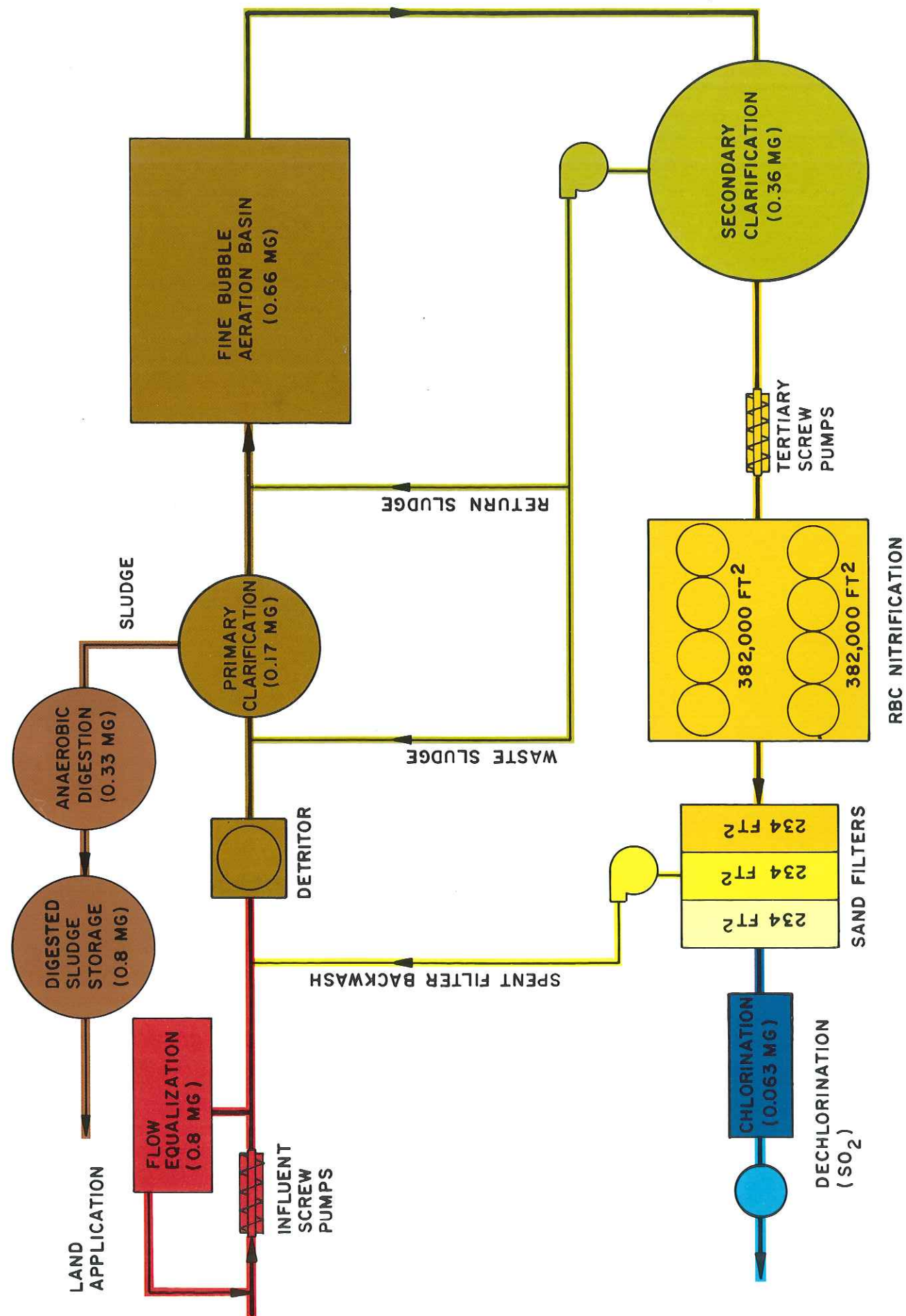
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DESIGN DATA

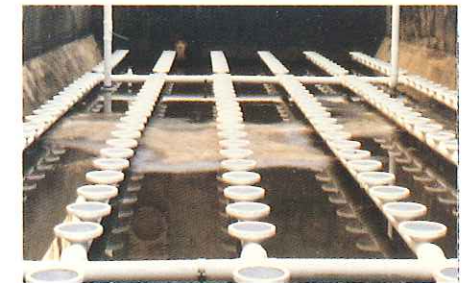
MINIMUM FLOW - 0.8 million gallons per day
 AVERAGE FLOW - 2.8 million gallons per day
 MAXIMUM FLOW - 4.5 million gallons per day
 DESIGN POPULATION - 11,000
 INFLUENT BIOCHEMICAL OXYGEN DEMAND - 180 mg/l
 INFLUENT SUSPENDED SOLIDS - 188 mg/l
 INFLUENT PHOSPHORUS - 7 mg/l
 INFLUENT AMMONIA NITROGEN - 20 mg/l
 EFFLUENT BIOCHEMICAL OXYGEN DEMAND - 6 mg/l
 EFFLUENT SUSPENDED SOLIDS - 10 mg/l
 EFFLUENT PHOSPHORUS - 0.5 mg/l
 EFFLUENT AMMONIA NITROGEN - 1.3 mg/l

AVERAGE PLANT REMOVAL

PRIMARY		
Biochemical Oxygen Demand	30%	
Suspended Solids	45%	
Phosphorus	10%	
Ammonia Nitrogen	—	
SECONDARY		
Biochemical Oxygen Demand	64%	
Suspended Solids	42%	
Phosphorus	70%	
Ammonia Nitrogen	—	
TERTIARY		
Biochemical Oxygen Demand	3%	
Suspended Solids	9%	
Phosphorus	—	
Ammonia Nitrogen	92.5%	
OVERALL PLANT		
Biochemical Oxygen Demand	96%	
Suspended Solids	96%	
Phosphorus	90%	
Ammonia Nitrogen	93%	

GLOSSARY OF TERMS

AERATION - The process which brings about contact between air and water by natural or mechanical means such as spray, bubbling or agitation.
AMMONIA NITROGEN - A relatively unstable form of nitrogen found in wastewater, which if left untreated, exerts an oxygen demand on the receiving stream.
BIOCHEMICAL OXYGEN DEMAND (BOD) - The amount of oxygen required by bacteria while stabilizing decomposable organic matter in the presence of air.
CLARIFIER - Tank or mechanical device for removal by gravity or suspended particles in liquid.
EFFLUENT - Partially or completely treated wastewater flowing out of a reservoir, tank or wastewater treatment plant.
INFLUENT - Raw or partially treated wastewater flowing into a reservoir, tank or wastewater treatment plant.
NITRATE NITROGEN - A relatively stable form of nitrogen formed by the bacterial oxidation of ammonia nitrogen.
PHOSPHORUS (P) - An element essential to the growth of plants that is found in relatively large quantities in raw wastewater.
SUSPENDED SOLIDS (SS) - Undissolved solid matter present in wastewater.



Mayor

Ronald Blanchard

Council Members

Virginia Gambrell

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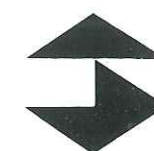
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Ayres, Lewis, Norris & May, Inc.

Engineers • Planners

Ann Arbor, Michigan

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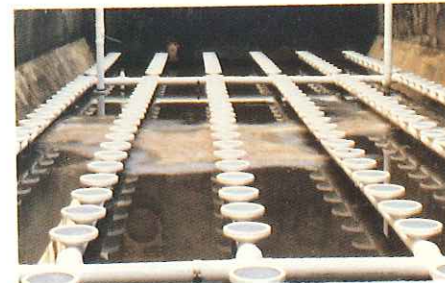
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